ECO 3271 Engineering Economics

Lecture-1
14.07.2026

Greek word "Oikonomia" থেকে Economics.

→ meaning ⇒ management of household

Germany

1776 → Adam Smith published a book named:

"An enquiry in the nature and the causes of the wealth of a nations".

According to Adam smith

Definition: Economics is the science which studies the nature and the causes of the ~~wealth~~ wealth of nation.

Lecture-2
21.07.2026

Chapter - 1

Definition and Subject matters of Economics

1

Year: 1776

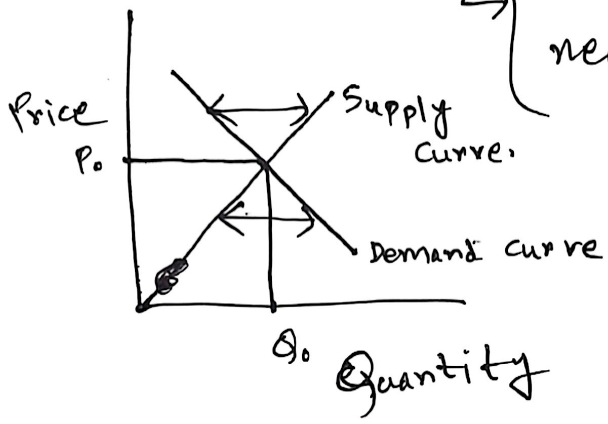
Author: Adam Smith

Book: "An enquiry in the nature and the causes of the wealth of nations."

Def: Economics is a science of wealth.

Features of Definition of Adam Smith:

1. Science of Wealth
2. Emphasis on production
3. Supports free marketing → Demand and Supply; Said govt interference is not needed in economy



Criticisms:

1. Ignores human welfare
2. Too materialistic definition → বস্তুগত
3. Narrow definition

2

Year: 1890

Author: Alfred Marshall

Book: "Principles of Economics"

Definition: "Economics is the study of mankind in the ordinary business of life."

Features and Criticism of definition of Alfred Marshall

Features: ① Emphasis on human welfare

② Wealth is only a means

③ Economics is a social science } man in society

Criticism: ① Welfare measurement problem

② Don't mention scarcity

③ Limited scope of his definition

③

Year: "1932" **Author:** "Lionel Robbins"

Book: "An Essay on the nature and significance of Economic Science"

Definition: Economics is a science which studies human behavior as a relationship between ends and scarce means which have a alternative uses

ends → unlimited human wants (অসীম)

scarce means → limited resource (দুষ্প্রাপ্য)

Features and Criticism of definition of Lionel Robbins;

Features: ① Scarcity is the central problem

② Resources have alternative uses.

③ Choice is necessary.

④ Efficient allocation of resource

Criticism: ① Don't mention technology

② Not suitable during unemployment.

* এগুলোর সব definition part. Ques. এ "Define Economics" বললে ~~mim~~ minimum দুইটা দিয়ে answer করতে হবে"

Subject matters of Economics

1. Production / Output: Production is the process of using resources like labor, capital and raw materials to make goods and services. This area helps us understand how businesses decide what to make and how much to produce

⇒ Focuses on transforming inputs into useful outputs to satisfy human needs.

2. Exchange: Exchange involves the buying and selling of goods and services between people, businesses or countries. This subject explains how prices are determined through the forces of supply and demand.

⇒ Usually takes place in markets using money as a medium of trade.

3. Distribution: Distribution deals with how total income and wealth are divided among individuals and factors of production. The goal is to understand how fairly or efficiently economic rewards reach different parts of society.

4. Consumption: Consumption is the act of using goods and services to satisfy immediate human wants and needs. It represents the final stage of economic activity and drivers overall economic demand. Studying consumption helps us to see how consumers choose to spend or save their money.

5. Employment: Employment focuses on how an economy uses its available labor for producing goods and services. It looks at job creation, working conditions, and the causes of unemployment. High employment is crucial because of generating income for individuals.

6. Economic Growth and Development: Economic growth refers to an increase in a country's total output of goods and services over time. Economic development goes a step further by including improvements in living standards, healthcare and education.

Microeconomics vs Macroeconomics

Lecture-3

23.07.2026

Features	Microeconomics	Macroeconomics
1. Definition	Mi. is a branch of economics which studies individual economic behavior (such as individual → consumers → producers → market etc	Mac. is a branch of economics which studies the economy as a whole (such as national output, national employment. Aggregate Demand and Supply, price level.
2. Unit of Analysis	Individual → consumers → produces → market	Aggregate Demand, Aggregate Supply.
3. Scope	Narrow	Broad.
4. Objectives	Efficient allocation of resources	Economic Stability.
5. Price	Individual commodities price	General Price level.
6. Income	Income of an individual	National income (GDP, GNP).
7. Market	Individual market.	Overall market economy
8. Policy Focus	Policy of single consumer produces	Fiscal. policy and monetary policy
9. Equilibrium	Partial equilibrium	General equilibrium.

10. Decision makers	Individual consumers " producers	Government, Central Bank (C.B.)

D) Central Economics Problem of Every Society:

1. What to produce –
   a) consumer goods/products.
   b) Capital goods (machinary) (money)
   c) services.

2. How to produce –
   a) Land (fixed)
   b) Labor
   c) Capital

   # if Capital > Labor
   → capital intensive production

   # if Labor > capital
   → Labor intensive production

3. Whom to produce:
   a) Rich
   b) Poor

Economics Systems

Lecture-4

28.07.2026

Economic System is the set of organizational arrangements and institutions that are established to solve economic problems.

Types:

① **Capitalist Economic System:** An economic ~~sy~~ system in which govt. does not control economic activity and ~~pro~~ all problems are solved by price mechanism is known as Capitalist Economic System. (C.E.S) Countries: USA, UK, Singapore.

Features of C.E.S:

① Private Property } Individual, businessman, company.

② Profit motive } maximization of profit

③ Market economy } $\underset{\text{Demand}}{D} = \underset{\text{supply}}{S} \Rightarrow$ equilibrium $\underset{\text{price}}{P}, \underset{\text{Quantity}}{Q}$

④ Competition.

⑤ Consumer sovereignty / freedom

⑥ Limited govt. role. } No govt. intervention here.

② Socialist Economic System: An economic system in which government decides how economic resources will be allocated and means of production are owned by the government. Countries: Soviet Union, Cuba, North Korea.

Features of S.E.S: ① Public Ownership } property, means

② Central Planning

③ No profit motive } social development

④ Equal distribution of income

⑤ Limited consumer choices.

⑥ Price is fixed by the ~~goat.~~ govt.

③ Mixed Economic System: An economic system in which most economic discussion result from the interacting of buyers and sellers in market but govt. play a significant role in the allocation of resources is known as M.E.S.

Features of M.E.S: ① Coexistance of public and private sectors.

② The state sometimes regulate ~~the~~ the economy

③ The govt. set development goals.

4. Consumer freedom.

5. The state works to minimize income inequility.

Quantity Demand: The amount of a good or service that a consumer is willing to and able to purchase at a given price.

economics এর ভাষায় বলে → ceteris paribus

Law of Demand: Holding everything else constant, when the price of a good or service falls, the quantity demanded of the good or service will increase and when the price of a good or service increases the quantity demanded decrease.

Demand schedule (চাহিদা সূচি): A table that shows the relationship between the price of a product and the quantity of the product is called Demand ~~st~~ Schedule.

Price table for printer

Price($)	Quantity
175	3
150	4
125	5
100	6
75	7

Table: Demand schedule.

Demand Curve: (চাহিদা রেখা)

A curve that shows the relationship between the price of a product and the quantity demanded of the product is called Demand Curve.

→ Demand curve কখনোই x/y অক্ষকে স্পর্শ করবে না।

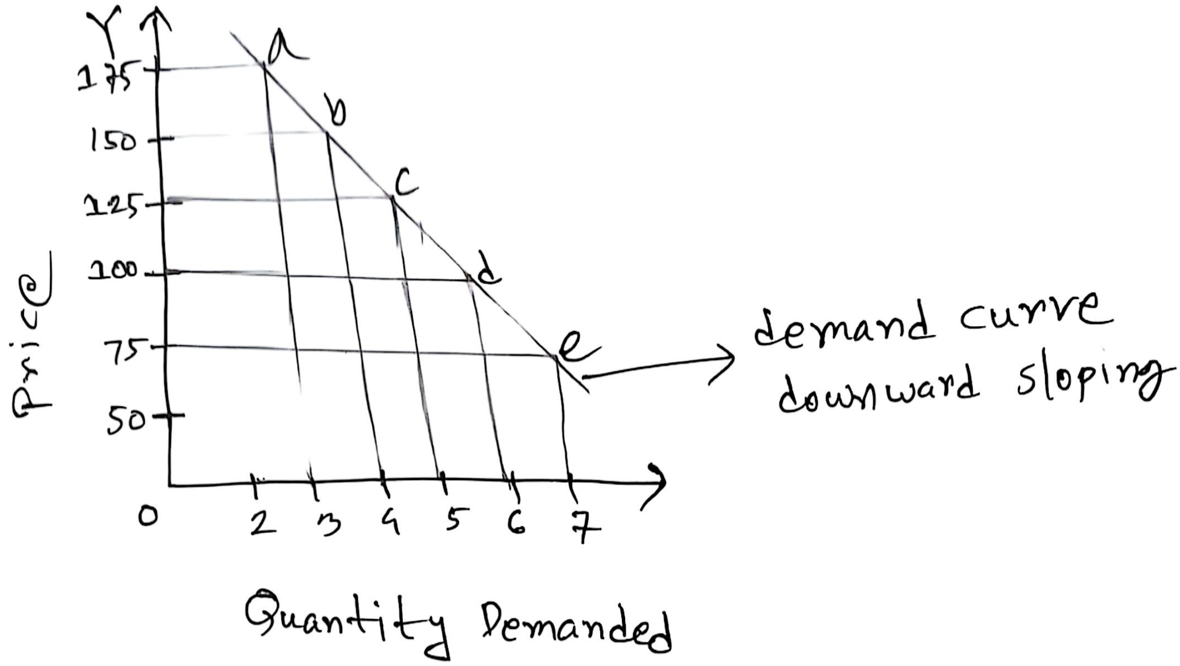


Figure: Demand curve

Demand equation: $Qd = a - bp$ → downward slopping

where,

Qd = Quantity Demanded

a = constant

$-b$ = constant that shows slope of demand curve

P = Price {যেগুলো থাকবে/ না থাকলে ধরে নিতে হবে}

Suppuse, Make a demand schedule where $Qd = 40 - 2p$

sol^n

P	Q
1	38
2	36
3	34

Variables that Shift Demand Curve

Those components on which the Quantity demanded depends except price.

① Income → If Income increases quantity demanded increases

② Taste ③ Population ④ Expected Future Price

~~④ Ex~~ ⑤ Price of related goods
- → substitute goods
- → complementary goods.

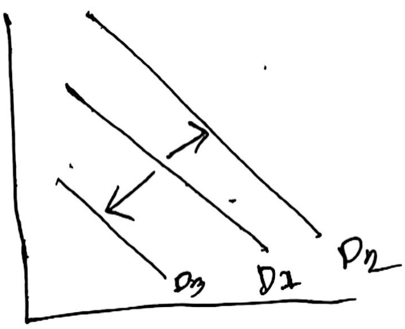


fig: shifting Demand Curve.

Lecture-6
06.08.2026

Supply / Quantity Supply

→ যোগান

Definition: The amount of good or service that a ~~firm~~ firm or producer willing and able to supply at a given price is called quantity supplied/supply.

Law of supply: Holding everything else constant, increases in price cause increases in quantity supplied and vice-versa, certeris paribus.

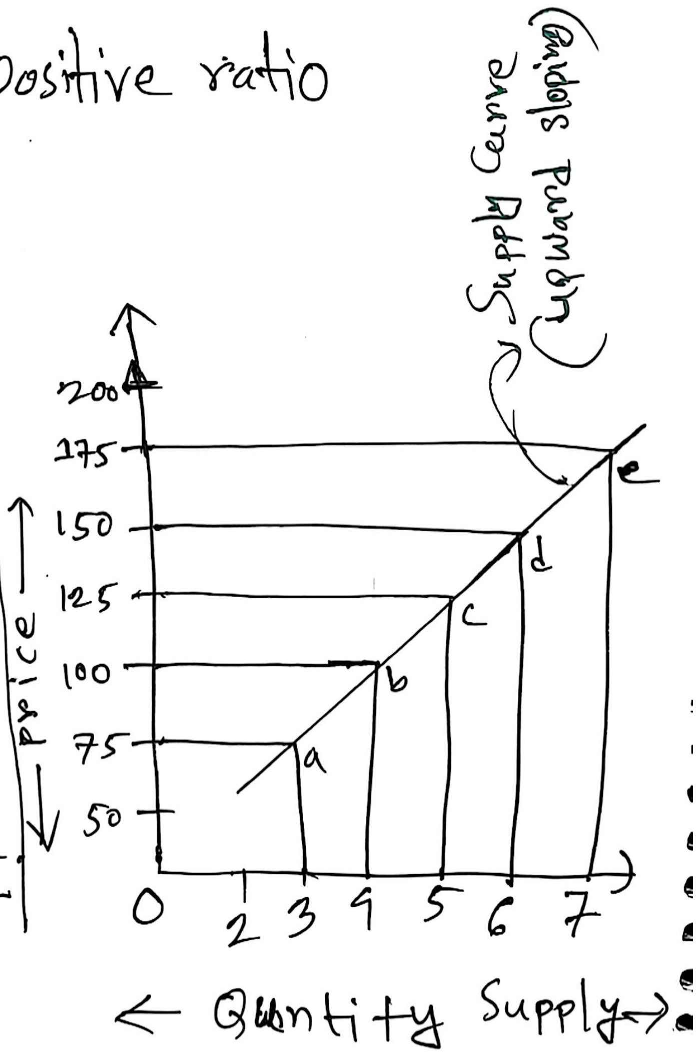
Supply schedule: The table that shows the relationship between the price of product and the quantity supplied of the product is called supply schedule:

Price	Quantity
175	7
150	6
125	5
100	6
75	7

→ positive ratio

Table: Supply Schedule

Supply Curve: A curve that show the relationship between the price of products and the quantity supplied of the product is called supply curve.



Supply equation:

$\boxed{Q_s = c + dp}$ — dependant ← Q_s; p → independent

or $\boxed{Q_s = -c + dp}$

P	Q
1	5
2	8
3	11
4	14

Here,

c = constant

d = constant show positive slope of supply curve

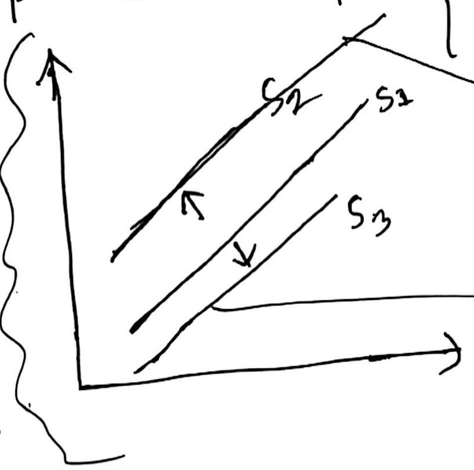
Q_0 = Quantity Supply

P = price

Variables that shift supply curve

① → price of input { Land, Labor, capital }

graph এরকম দেখতে হবে।



→ decrease in supply
(উপরের দিকে ডানে সরবে)

→ Increases in Supply
(নিচের দিকে বামে)

② Technological change

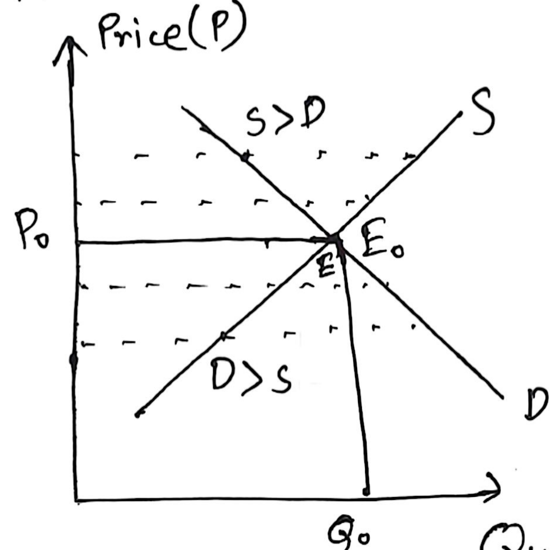
③ Expected future price

④ Number of firms in the market.

Market Equilibrium (বাজার ভারসাম্য)

Market equilibrium is a situation in which quantity demanded equal ~~e~~quantity supplied.

Def - Grap - Description
(উপরে) (মাঝে) (নিচে) (exam-এ)



$P_0 = Q_0$ হলে market equilibrium হবে।

A market equilibrium graph plots Price on the vertical Y-axis against Quantity (Q) on the horizontal X-axis. It features a downward-sloping Deman Curve (D) and an upward-sloping Supply Curve (S) which intersect Equilibrium Point (E). This intersection determines the Equilibrium Price (P_0) and Equilibrium Quantity (Q_0), where quantity demanded equals quantity supplied ($Q_D = Q_s$) and the market is balanced. Any price above P_0 creates a surplus (excess supply) that forces prices down, while any price below P_0 causes a shortage (excess demand) that pushes prices back up to equilibrium.

Lecture-7

11.08.2026

Deman equation

$Qd = a - bp$

Supply equation

$Qs = c + dp$

$\Rightarrow Qs = -c + dp$

Question-1: $Qd = 80 - 3P$ / $Qs = -10 + 2P$; Find the equilibrium price and quantity from the above equations.

Soln: Given that, $Qd = 80 - 3P$ ---- (i) $Qs = -10 + 2P$ ... (ii)

We Know, in equilibrium condition,

$$Qd = Qs$$

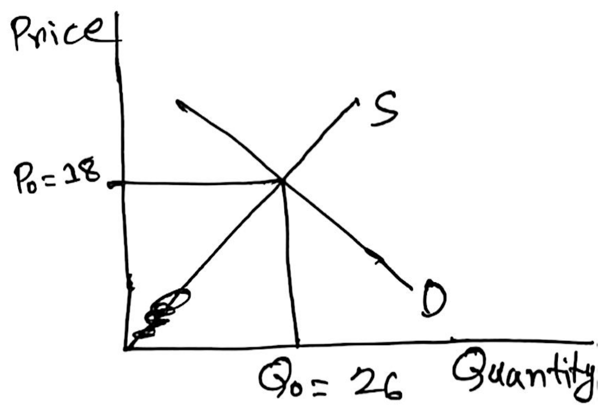
$$80 - 3p = -10 + 2p \Rightarrow 2p + 3p = 80 + 10$$

$$\therefore P = 18 \text{ ---- (iii)}$$

Putting $p = 18$ in equation (i),

we get, $Qd = 80 - 3 \times 18$

$Qd = 26$



So the equilibrium price = 18

and Quantity = 26

Question-2: $Q_d = 1200 - 10$ | $Q_s = -400 + 6p$

Find equilibrium price and quantity from the above equation.

Ans: Given that,

$$Q_d = 1200 - 10p \cdots (i) \quad | \quad Q_s = -400 + 6p \cdots (ii)$$

We know, in equilibrium condition,

$$Q_d = Q_s$$

$$1200 - 10p = -400 + 6p$$

$$\Rightarrow 16p = 1600$$

$$\therefore p = 100 \cdots (iii)$$

Putting $p = 100$ in equation -- (ii)

we get, $Q_s = -400 + 6 \times 100$

$$= 200$$

So the equilibrium price = 100

and Quantity = 200

Elasticity (স্থিতিস্থাপকতা)

Responsiveness of dependent variable due to change in independent variable.

Elasticity of Demand

~~Percentage change in quantity demanded due to percentage in price.~~

P	Q
2	3
5	4
7	20

Types:

(i) **Price Elasticity of demand:** Percentage change in quantity demanded due to percentage change in price. Mathematically,

$$\boxed{\frac{\Delta Q}{\Delta P} \times \frac{P}{Q}}$$

(ii) **Income elasticity of Demand:** Percentage change in quantity demanded due to percentage change in income. Mathematically,

$$\boxed{\frac{\Delta Q}{QI} \times \frac{I}{Q}}$$

(iii) **Cross Price Elasticity:** Percentage change in quantity demanded of one goods due to percentage change in price of another goods.

Elasticity of Demand Curves

(i)

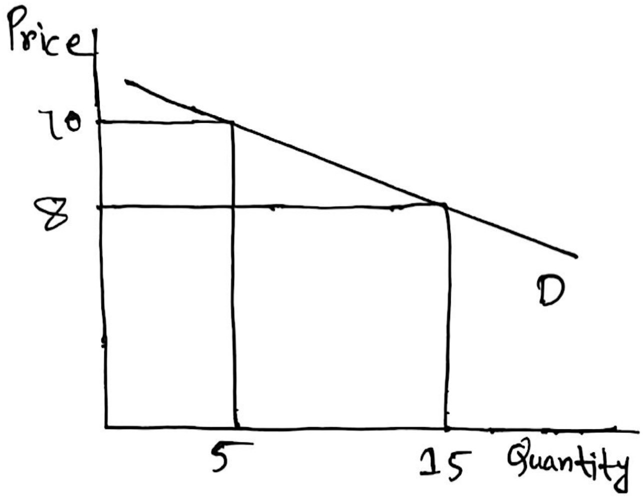


Fig: Elastic Demand ($E_d > 1$)

(ii)

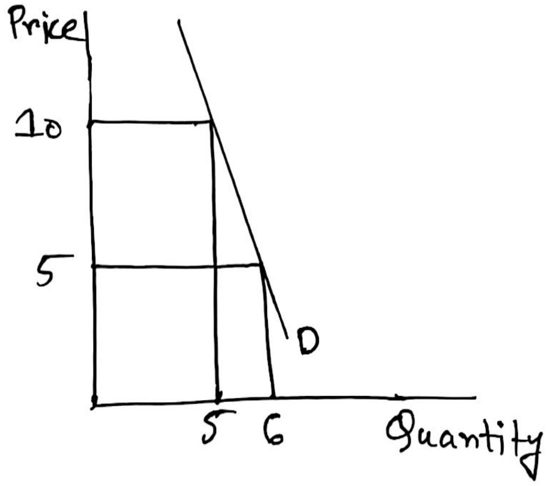


Fig: Inelastic demand ($E_d < 1$)

(iii)

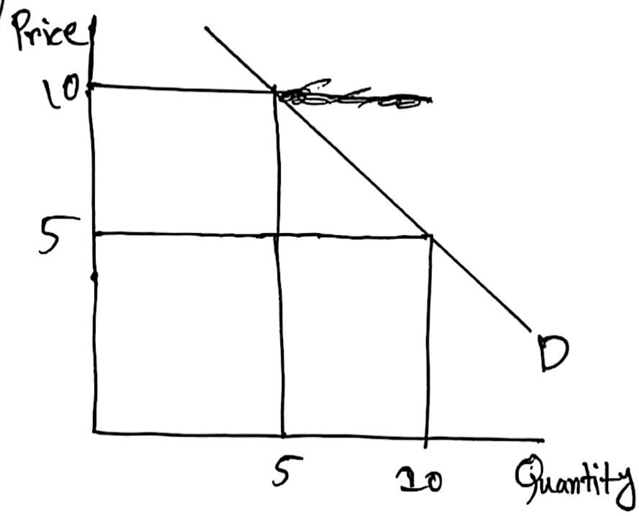


Fig: Unit Elasticity ($E_d = 1$)

(iv)

Fig: Zero Elasticity ($E_d = 0$)

(v)

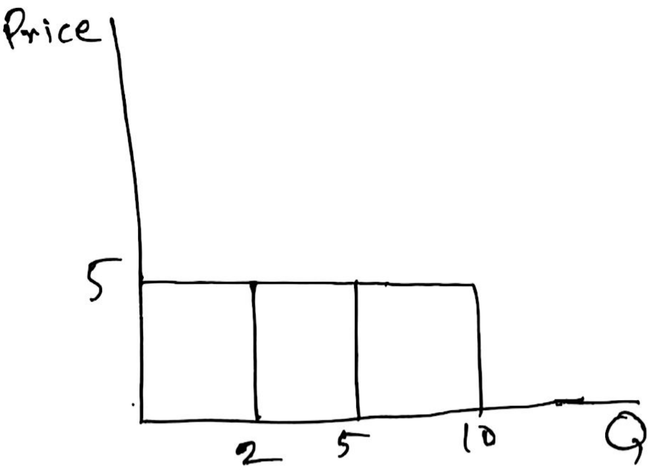


Fig: Infinite elasticity ($E_d = \infty$)

Elasticity of Supply

Lecture-8

13.08.2026

Percentage change in quantity supplied due to percentage change in Price, Mathematically, $E_s = \frac{\Delta Q_s}{\Delta P} \times \frac{P}{Q_s}$

i) Elastic Supply:

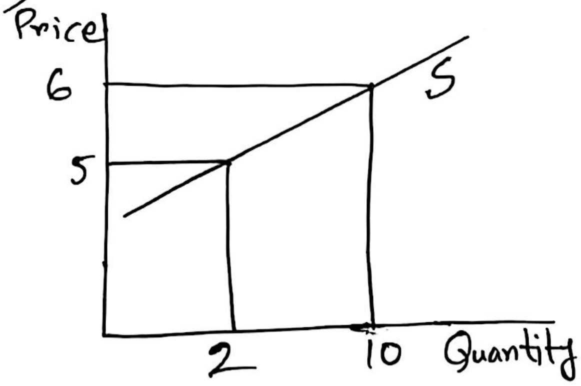


Fig: Elastic Supply ($E_s > 1$)

ii) Inelastic

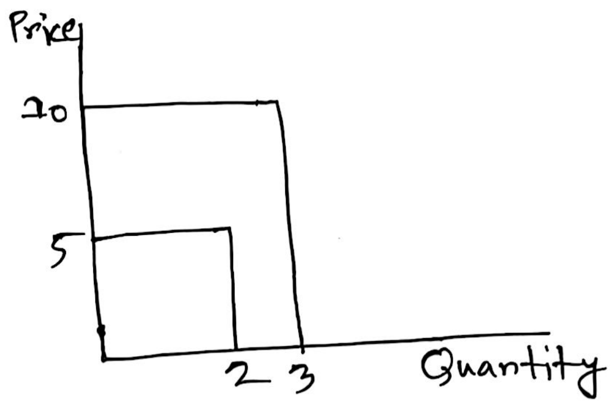


Fig: Inelastic Supply ($E_s < 1$)

iii) Unit Elasticity

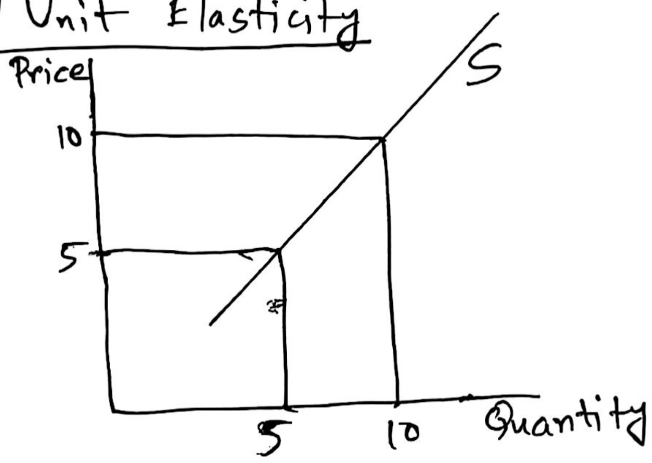


Fig: Unit Elasticity ($E_s = 1$)

iv) Zero Elasticity:

→ Perfectly Inelastic

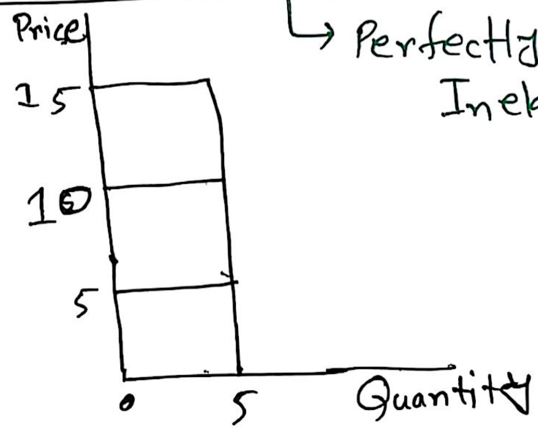


Fig: Zero elasticity ($E_s = 0$)

v) Infinite

→ Perfectly elastic

Fig: Infinite elasticity ($E_s = \infty$)

Chapter-3

Theory of Firm

Production (উৎপাদন): Production is a process that involves transformation of inputs such as land, labor, capital and other raw materials into output such as goods and services.

Production function (উৎপাদন অপেক্ষক): Production function is a function which shows the technical relationship between inputs and outputs.. Symbolically,

$$Q = f(L, L_b, K, R, T, t)$$

where,

L = Land

L_b = Labor

K = Capital

R = raw materials

T = Technology

t = time

Factors of Production (উৎপাদনের উপাদান): There are mainly four types of factors of production:

(i) Land

(ii) Labor

(iii) Capital

(iv) Entrepreneur skill

Factors of Production

Lecture-9

18.08.2026

(i) **Land:** Land refers to the surface of land and also all gifts of nature such as rivers, oceans, climate, fisheries, forests.

(ii) **Labor:** Labor refers to all mental and physical work undertaken for some monetary unit.

(iii) **Capital:** Capital means all man made resources.

(iv) **Entrepreneurs skill:**

Organization: Org. refers to the services of an entrepreneur who controls and manages the policy of firms and undertaken all possible risk.

Cost of production

Total cost: The sum of all cost for different factors of production is called total cost.

$$TC = Q \times P$$

Average cost: Total cost divided by output

$$AC = \frac{TC}{Q}$$

~~Marginal cost: Total cost divided by output~~ A

Marginal cost: Additional cost for producing additional unit of output.

$$MC = \frac{\Delta TC}{\Delta Q}$$

Lecture-10	
20.08.2026	

Total cost is divided into two ~~categories~~ categories.

i) Total fixed cost (TFC)

ii) Total variable cost (TVC)

Total Fixed Cost: TFC are those expenses which do not changes with the change in output/production. For example - Rent, ~~Interest~~, Interest of capital, wages of permanent worker etc.

Total Variable Cost: TVC are those expenses which change with the change in output/production. For example - cost of raw materials, wage of temporary labor, transport cost etc.

Cost Curves: Cost Curves are the curves that shows the relationship between production and production cost.

Here, $\boxed{\text{Total cost} = TFC + TVC}$